

MECHANIC/ELECTRICIAN

All the maintenance operations assigned by the Maintenance Shift Supervisor shall be performed in accordance with the safety regulations set out and with the suitable protective equipment for this job.

The actions shall be carried out with the highest degree of professionalism and be assessed to prevent them from happening again, with the purpose of suggesting new actions and reaching a greater efficiency of the machinery.

RESPONSIBILITIES

- Implement the preventive and corrective maintenance tasks in order to reach the highest possible efficiency of the machinery, reducing the production line downtime.
- Meet the safety regulations set out in each maintenance action.

FUNCTIONS

- Solve the mechanical / electric issues, when required, which might stop the proper functioning of the production line, trying to fix it as soon as possible (also considering the complete repair in maintenance after production).
- Perform maintenance tasks (inspection routes, lubrication, scheduled repairs and repairs in the production line) according to the indications given by the maintenance/production shift supervisor.
- Report any anomalies in the machinery and equipment to their maintenance/production shift supervisor.
- Make regulations specialized in the machinery and equipment.
- Fill in the daily maintenance sheet.
- Cooperate in the proper supply of spare parts in the stockyard.
- Suggest improvements and innovations to increase the efficiency, the quality and the productivity.

PROTECTIVE EQUIPMENT

- Safety gloves.
- Safety boots.
- Noise-abatement ear plugs or headset.

EXTRUDER MECHANIC

All the maintenance operations indicated by their supervisor shall be performed, meeting the safety regulations set out and using the proper protection means suitable for such work.

The actions shall be carried out with the highest degree of professionalism and be assessed to propose actions to prevent them from happening again, with the purpose of a greater efficiency of the machinery.

The specialized mechanic in the extruder machine shall supervise the state of this machine with a thorough daily control of all its elements: daily adjustments and wear of all the parts, the wear of rollers and slippers, bench guides, pushing chains and the tile cutting system.

RESPONSIBILITIES

- Undertake the preventive and corrective maintenance tasks with the purpose of a greater efficiency, reducing downtime in the production line.
- Keep the daily maintenance tasks in the extruder machine in order to reach the tile production quality set out, meeting the concrete roof tile regulations in the (..... Bureau of Standards.)
- Meet the safety regulations set out in each maintenance action / operation.

FUNCTIONS

- Adjustments and control of the extruder machine, the daily wear of all the parts in the machine, the evolution and wear of rollers and slippers, bench guides, pushing chain guides, the pushing chain, chain tensioners, sprockets, trolleys, pushing fingers and the cutting system (blades, guides, rolling bearing and blade cleaning nozzles).
- Solve the problems which stop the proper functioning of the line.
- Report any anomalies in the machinery to their maintenance shift supervisor.
- Fill out the extruder machine maintenance sheet and the daily sheet.
- Cooperate in the proper supply of the spare parts in the warehouse.
- Suggest suitable improvements and innovations in order to keep and increase the level of quality and productivity.

PROTECTIVE EQUIPMENT

- Safety gloves.
- Safety boots.
- Noise-abatement ear plugs or headset.



Or similar.

EXTRUDER OPERATOR

These are the instructions to ensure the proper tile formation which fulfils the quality requirements set out.

RESPONSIBILITIES

- Ensure the proper tile formation.
- Supervise the surface appearance, the cutting and the punching.
- Control the release agent application.

DESCRIPTION

Before production.

- Review the mould inverter.
- Check the filling level of the release agent application tank.
- Check the proper functioning of the oil application.
- Check the side scrapers on both sides of moulds.
- Check that the head on the extruder is suitable for the production set out.
- Check that the tile cutting blades are properly adjusted to the mould profile and to the fixing hole nails.
- Check the proper functioning of the lubrication application on the base of the mould profiles.
- Put the extruder head hopper, oil the hopper and all the parts of the machine with a brush and release agent.
- Oil the moulds which may remain from the oil application to the extruder.
- Prepare the proper mould supply in the input conveyor to the extruder in order to start.
- Report any problem or need to the Shift Supervisor.
- If everything is fine, the Extruder is ready.

At the beginning of production.

- Fill up the extruder head hopper with concrete.

Once the above operations have been performed, press the start button in the automatic production line checking that:

- The moisture of the mixture is suitable for the formation of the whole tile (through the tile surface appearance and the texture we can determine whether it is properly formed).
- The release agent application is correct: when the concrete can be easily separated from the mould without any accumulation of release agent in the mould supporting ribs:

- For which it is required to remove one or two moulds at the beginning, in order to validate the release agent application.
- The weight of wet tiles is correct in accordance with the tolerances set out.
 - For which it is required to remove one tile from the production line checking the weight of the tile. If in accordance with the tolerances set out, production shall go on.

(If the weight of the tile is out of the tolerances, production shall be stopped and the incident shall be reported to the shift supervisor so as to correct it).

The concrete removed from the tile in this operation shall be put into the container for this purpose.

- The dimensions and the wet tile punching are correct when in accordance with the tolerances set out:
 - For which it is required to remove one tile from the production line to check all the dimensions. If in accordance with the tolerances set out, production shall go on.

(If any dimensions are out of the tolerances, production shall be stopped and the incident shall be reported to the shift supervisor so as to correct it).

The concrete removed from the tile in the operation shall be put into the container for this purpose.

**** All the moulds removed from the production line shall be returned to the production line or replaced by others so as not to decrease the number of moulds in the circuit.**

During production.

Tile.

- Check the supply, compaction, release agent application, weights and dimensions of the tiles (one tile per hour).
- Supervise the effort of the pushing chain and the compaction in the tiles.

Production line.

- Supervise the proper functioning of the machinery.
- Stop the production line in order to correct incidents when:
 - The cut is wrong.
 - Lack of supply or compaction in tiles.
 - The concrete moisture is wrong.
 - The pushing chain works incorrectly.

- Any anomalies in the functioning of the machines.

(Being noted in the extruder control sheet.)

- Ring the bell to alert the mixer operator of any incidents in that area, in the production line or any help needed.
- Help maintenance in the head change.

****** In the event of stops in the production line, assistance must be provided in order to solve the stop as soon as possible. If assistance is not needed, clean the assigned work area.

The moisture of the mix.

- Stay in constant contact with the mixer operator, in case it is required to vary the moisture of the mix (wrong moisture) or to adjust the moisture if a remote adjusting device from the extruder is provided.

Mould.

- Remove the moulds stacked in the production line on their way to the extruder or tiles / pieces of tiles which have remained on the mould in the demolding machine.
(If it is not removed, it shall stop the production line.)
- Review all the upper and lower side parts of a mould (at least one mould a day) in order to detect any dirt, erosion, mark or wear and note it in the extruder control sheet.

After production.

- Put the extruder area in manual mode.
- Empty the extruder head hopper.
- Clean the work area carrying out the following operations:
 - Unplug the control panel and press the emergency stop button or safety equipment to ensure safe work.
 - Extract the extruder head, lay it on the transportation trolley and clean the entire extruder.
 - Clean the extruder unit according to the cleaning sheet.
 - Clean the assigned work area.
- Fill out the Extruder Control Sheet and the Stop/Incident Sheet for preventive maintenance.

MEANS AND WORK TOOLS

- Weighing and measurement tables.
- Tile weighing scale.
- Calipers to measure thickness, splints and calibration.
- Small digital depth gauge.
- Digital caliper of 150 mm or 200 mm.
- Rectangular calibrated splint of 200x30x6 mm.
- Round calibrated splint of 200x10 mm diameter.
- Air compressed feed line.
- Low flow air gun.

PROTECTIVE EQUIPMENT

- Safety gloves.
- Safety boots.
- Noise-abatement ear plugs or headset.
- Safety glasses for cleaning and washing the extruder head, the pushing unit and the cutting unit.

NECESSARY TRAINING

Before performing this job, the operator must be trained to:

- Identify the usual wet tile colors and assess the concrete plasticity to make tiles.
- Identify and use the measurement tools to check the dimensions of tiles.
- Identify the fixing and adjusting elements of the extruder head.
- Use all the manual and automatic control panels of the extruder area.

In any case, before taking this position, they shall spend at least one week along with another operator who has performed the job satisfactorily for 6 months according to the Shift Supervisor.